Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the accuracy during training.

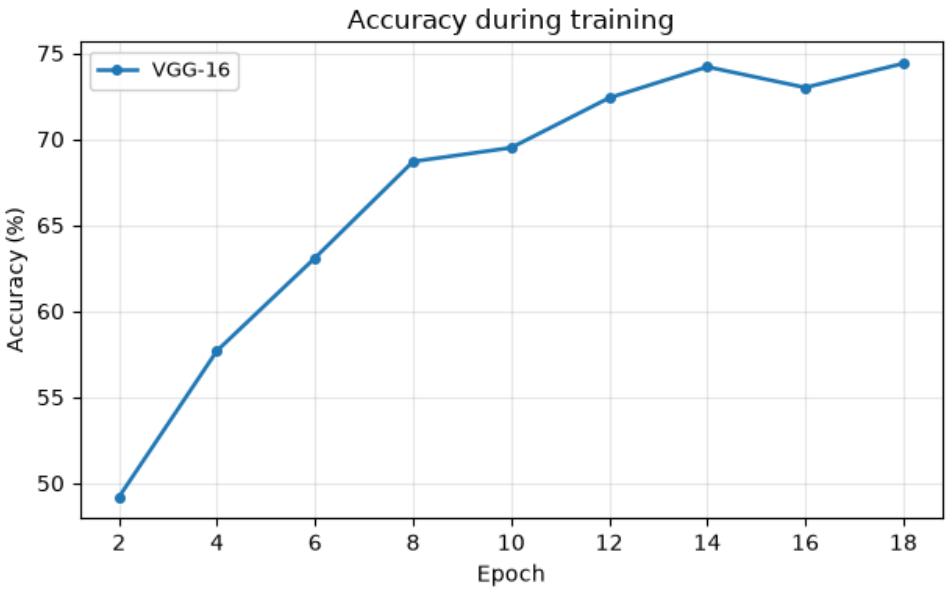


Figure 1: Accuracy during training for the models in this experiment.

As shown in Figure 1, VGG-16 scores 11.8 points higher on 10 than on 4. As shown in Figure 1, VGG-16 scores 11.0 points higher on 14 than on 4. VGG-16 reaches 78.5% accuracy at epoch 16. At epoch 12, VGG-16 achieves 71.0% accuracy.

Figure 2 shows the test accuracy by dataset.

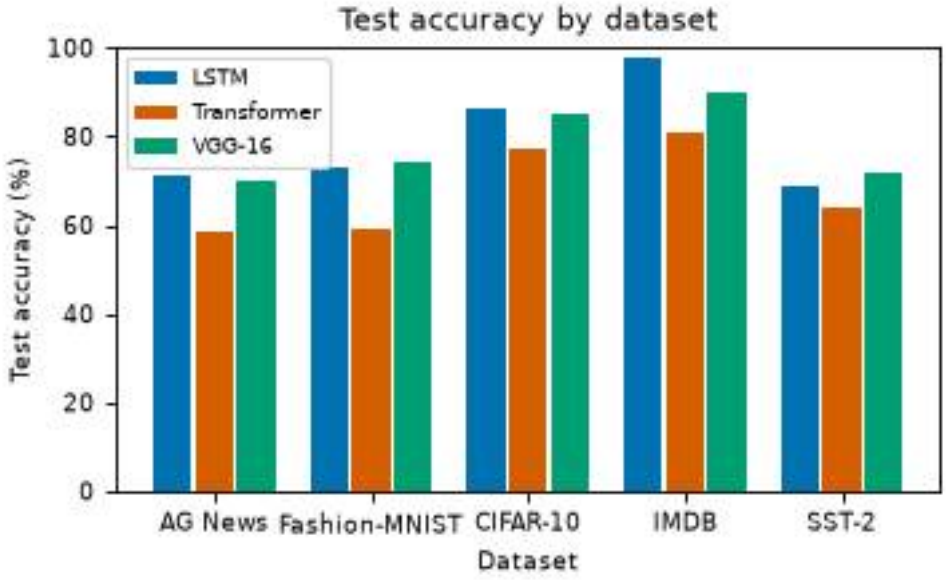


Figure 2: Test accuracy by dataset for the models in this experiment.

LSTM outperforms VGG-16 by 11.2 percentage points on AG News. LSTM achieves 71.6% test accuracy on SST-2 (Figure 2).

3 Discussion

Overall, the results are in line with what we expected from the literature.